

BRYAN HILTON

Mechanical Engineering Undergrad at BYU

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OBJECTIVE

Driven student and experimentalist with exceptional hands-on research experience. Seeking both local and foreign opportunities to collaborate with a team tackling challenging problems in the nuclear energy sector.

EDUCATION

Brigham Young University: Mechanical Engineering Major

Expected Graduation: April 2028

Current total credits: 137, Major GPA: 3.80

SKILLS

- High Vacuum Systems
- Monte Carlo Simulation
- CAD Modeling and Drafting
- High Voltage Systems
- Python (numpy), C, C++, HTML
- Manual Machining, 3d Printing
- Arduino, Raspberry pi
- Measurement Electronics
- Excel: Data Analysis

EXPERIENCE

Research Assistant

Sep. 2026 – present

- Thermal analysis of SiC/SiC composites for nuclear fuel cladding applications
- Laser Flash Analysis and mass data collection

American Nuclear Society Chapter Leadership

April 2026 – present

- Designing and building a python-based interactive MC neutron transport simulation
- Scheduling & organizing activities for fellow BYU students

Vacuum Technician for BYU Physics Department

July 2025 – June 2026

- Designed, assembled, and repaired many high vacuum systems, pumps, and gauges
- Built high voltage power supplies, repaired and calibrated analog electronics

BYU Rocketry Club / Aeronautics Club

2021 – 2025

- Used CAD, 3D printing, simulation, helped design payload delivery mechanism for the DBF team

Robotics Coach at Boys & Girls Club

Summer 2021

- Taught a youth class how to build and program their own robots
- Hosted a competition to allow them to test their designs

Other Project Experiences:

- Designed & built a high vacuum system, high voltage supply, and computer simulations to investigate applications of surface wakefield acceleration in proton therapy.
- Built robots with LIDAR & raspberry pi, remote control and sensors with arduino, served as high school robotics club president. Tracked near-Earth asteroids and built computer simulations to predict trajectories for the MIT Summer Science Program in Astrophysics.

PERSONAL BIO

I grew up in New Mexico, as the 2nd oldest of 10, loving all things outdoors, hiking, running, and tennis. While a volunteer English teacher in Mongolia for two years, I gained a deeper love and respect for others that has continually motivated me to make the world a better place. I have always been fascinated by many areas of physics, but nuclear power has stood out to me particularly as an important part of the solution to increasing global energy needs. I hope to help implement safe and effective energy solutions at home and abroad.



(link to my personal website)